Week 3: Seller and Inventory Management System

Scenario:

The e-commerce company wants to maintain information about customized gifts and their categories.

Each product belongs to a specific category, and the system manages product details such as **product name, price, and stock availability**.

The main categories include **Birthday Party, Anniversary, Baby Shower, Wedding, Housewarming, and Graduation**.

1.create Seller and Inventory Tables

```
SQL> CREATE TABLE Seller(
  2      Seller_ID VARCHAR2(10),
  3      Seller_Name VARCHAR2(25),
  4      Phone NUMBER(10),
  5      Address VARCHAR2(50)
  6  );

Table created.
```

```
SQL> CREATE TABLE Inventory(
  2      Inventory_ID VARCHAR2(7),
  3      Product_ID VARCHAR2(7),
  4      Seller_ID VARCHAR2(7),
  5      Stock_Quantity NUMBER(11),
  6      Stock_Status VARCHAR2(11),
  7      Last_Updated DATE
  8  );

Table created.
```

2. Establish Relationships

Seller and Product

One seller can supply multiple products.

One product can be supplied by one or multiple sellers.

Relationship:

Seller → Product is One-to-Many

Product and Inventory

Each product has inventory information.

Inventory tracks available stock.

Relationship:

Product → Inventory is One-to-One

Primary Key added for Both Tables:

```
SQL> ALTER TABLE Seller ADD CONSTRAINT PK_Seller PRIMARY KEY(Sel
ler_ID);

Table altered.
```

```
SQL> ALTER TABLE Inventory  ADD CONSTRAINT PK_Inv PRIMARY KEY(In
ventory_ID);

Table altered.
```

Foreign Key for Inventory Table:

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT FK_Seller FOREIGN KEY(
Seller_ID) REFERENCES Seller(Seller_ID);

Table altered.
```

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT FK_Prod FOREIGN KEY(Pr
oduct_ID) REFERENCES Product(Product_ID);

Table altered.
```

Constraints to make Unique Email ID: To make Email NOT NULL

```
SQL> ALTER TABLE Seller ADD CONSTRAINT Un_mail UNIQUE(Email);

Table altered.

SQL> ALTER TABLE Seller MODIFY Email NOT NULL;

Table altered.
```

Add Constraint to Check the Stock Quantity is Positive

```
SQL> ALTER TABLE Inventory
  2  ADD CONSTRAINT CK_Quan CHECK(Stock_Quantity >= 0);

Table altered.
```

Insert Values to Seller Table

```
SQL> INSERT INTO Seller VALUES('S00007','Kumar Traders',9876543210,'kumart@gma
il.com','Anna Nagar, Chennai');

1 row created.

SQL> INSERT INTO Seller VALUES('S00008','Green Mart',9123456780,'greenmart@gma
il.com','RS Puram, Coimbatore');

1 row created.

SQL> INSERT INTO Seller VALUES('S00009','Sri Lakshmi Stores',9988776655,'srila
kshmi@gmail.com','Gandhipuram, Coimbatore');

1 row created.

SQL> INSERT INTO Seller VALUES('S00010','Tech World',9090901234,'techworld@gma
il.com','T Nagar, Chennai');

1 row created.

SQL> INSERT INTO Seller VALUES('S00011','Fashion Hub',9345678123,'fashionhub@g
mail.com','KK Nagar, Madurai');

1 row created.

SQL> INSERT INTO Seller VALUES('S00012','City Electronics',8765432190,'cityele
ctronics@gmail.com','Trichy Road, Trichy');

1 row created.
```

```
SQL> SELECT Seller_ID FROM Seller;

SELLER_ID
----------
S00007
S00008
S00009
S00010
S00011
S00012

6 rows selected.
```

Insert Values to Inventory Table

```
SQL> INSERT INTO Inventory VALUES('Inv005','P001','S00007',5,'Available',TO_DA
TE('2026-09-16','YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO Inventory VALUES('Inv006','P002','S00008',2,'Available',TO_DA
TE('2026-09-17','YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO Inventory VALUES('Inv007','P003','S00009',8,'Available',TO_DA
TE('2026-09-18','YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO Inventory VALUES('Inv008','P004','S00010',0,'Unavailable',TO_
DATE('2026-09-19','YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO Inventory VALUES('Inv009','P005','S00011',6,'Available',TO_DA
TE('2026-09-20','YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO Inventory VALUES('Inv010','P006','S00012',3,'Available',TO_DA
TE('2026-09-21','YYYY-MM-DD'));

1 row created.
```

```
SQL> SELECT * FROM Inventory;

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
Inv005  P001    S00007               5 Available   16-SEP-26
Inv006  P002    S00008               2 Available   17-SEP-26
Inv007  P003    S00009               8 Available   18-SEP-26
Inv008  P004    S00010               0 Unavailable 19-SEP-26
Inv009  P005    S00011               6 Available   20-SEP-26
Inv010  P006    S00012               3 Available   21-SEP-26

6 rows selected.
```

Report 1: Seller-wise Product Report

Display:

- Seller name
- Number of products supplied
- Available stock

```
     SELECT s.Seller_Name, COUNT(i.Product_ID) AS Number_of_Products, SUM(i.St
ock_Quantity) AS Available_Stock FROM Seller s JOIN Inventory i ON s.Seller_ID
 = i.Seller_ID WHERE i.Stock_Status = 'Available' GROUP BY s.Seller_Name;

SELLER_NAME               NUMBER_OF_PRODUCTS AVAILABLE_STOCK
------------------------- ------------------ ---------------
Green Mart                                 1               2
City Electronics                           1               3
Sri Lakshmi Stores                         1               8
Fashion Hub                                1               6
Kumar Traders                              1               5

SQL>
```

Report 2: Stock Availability Report

Display:

- Product name
- Stock quantity
- Stock status

```
SQL> SELECT p.Product_Name, i.Stock_Quantity, i.Stock_Status FROM Product p JO
IN Inventory i ON p.Product_ID = i.Product_ID;

PRODUCT_NAME
--------------------------------------------------------------------------------
--
STOCK_QUANTITY STOCK_STATU
-------------- -----------
Laptop
             5 Available

Smartphone
             2 Available

Headphones
             8 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
--
STOCK_QUANTITY STOCK_STATU
-------------- -----------
Smart Watch
             0 Unavailable

Keyboard
             6 Available

Mouse
```

Report 3: Inventory Analysis

Find: Total products available

```
SQL> SELECT COUNT(*) AS TOTAL_AVAILABLE_PRODUCTS FROM Inventory WHERE STOCK_ST
ATUS = 'Available';

TOTAL_AVAILABLE_PRODUCTS
------------------------
                       5
```

Products out of stock :

```
SQL> SELECT p.Product_ID, p.Product_Name, i.Stock_Quantity, i.Stock_Status FRO
M Product p JOIN Inventory i ON p.Product_ID = i.Product_ID WHERE i.Stock_Quan
tity = 0;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
--
STOCK_QUANTITY STOCK_STATU
-------------- -----------
P004
Smart Watch
             0 Unavailable
```

Highest stocked products :

```
SQL> SELECT p.Product_ID, p.Product_Name, i.Stock_Quantity FROM Product p JOIN
 Inventory i ON p.Product_ID = i.Product_ID WHERE i.Stock_Quantity = (SELECT M
AX(Stock_Quantity) FROM Inventory);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
--
STOCK_QUANTITY
--------------
P003
Headphones
             8
```

Average inventory quantity:

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory_Quantity FROM Inventory;

AVERAGE_INVENTORY_QUANTITY
--------------------------
                         4

SQL>
```